B-trees

Part V Advanced Data Structures

Scope

Packing multiple keys into a page reduces disk-page search paths. Insertion and deletion must move child pointers with keys to preserve global order, occupancy bounds, and equal leaf depth. Page I/O and within-page CPU work are counted separately.

Prerequisites

Multiway search trees, disk I/O, and height balance.

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Chapter route

Learning objectives

- Define node capacity using minimum degree t.
- Analyze search, splits, and merges in page I/O.
- Check root and key-count boundaries.
- 18.1 Definition of B-trees
- 18.2 Basic operations on B-trees
- 18.3 Deleting a key from a B-tree

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Definition of B-trees

A node stores multiple sorted keys and child pointers, typically in one disk page. Keys are distinct; associated data or record pointers move with each key. Duplicate insertion reports EXISTS. This B-tree stores data in internal nodes, unlike a B+ tree with leaf-only data and copied parent separators.

Minimum degree t is an integer≥ 2. Every nonroot holds $t-1..2t-1$ keys; a nonempty root holds$1..2t-1$. An internal node with k keys has exactly k+1 children. All leaves have depth h, with root depth0.

Keys $a0<\ldots<a_{(k-1)}$ partition global child ranges: child0 keys<a0, child i keys in$(a_{(i-1)},a_i)$, child k keys>$a_{(k-1)}$. Sorted keys within each node alone do not prove these subtree bounds.

An empty tree is a zero-key leaf root at height0. The height theorem below assumes n≥ 1.

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Upper and lower height bounds

A minimum-occupancy nonempty height-h tree has one root key and two children; other nodes have t−1 keys. Depth i≥1 contains at least2t^(i−1) nodes, so

$$n \geq 1+\Sigma_{i=1}^{h} 2t^{i-1}(t-1)=2t^h-1,$$

hence $h \leq \log_t((n+1)/2)$. At h=0 the sum is empty and the formula still holds.

Conversely, at most2t−1 keys and2t children per node give

$$n \leq (2t-1)\Sigma_{i=0}^{h}(2t)^i=(2t)^{h+1}-1,$$

hence $h \geq \log_{(2t)}(n+1)-1$. These describe sparsest and densest legal trees; they do not predict a typical occupancy.

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Page bounds for a billion keys

For t=100,n=10^9, height2 holds at most$200^3-1=7,999,999$ keys, insufficient. Height3 holds up to$200^4-1=1,599,999,999$, so minimum possible height is3.

Height4 needs at least$2\cdot 100^4-1=199,999,999$ keys and remains possible. Height5 needs19,999,999,999,too many. Thus $h\in\{3,4\}$,giving4 or5 visited nodes.

With a resident root and no other cached pages, this is3 or4 page reads; without cached root,4 or5. Linear within-page search needs O(t) comparisons,CPU O(th). Binary search reduces search comparisons to O(h log t),but insertion/deletion array movement may still cost O(t) per level.

Page size,key/record/child-pointer sizes,and metadata determine t. I/O counts are not measured disk latencies and cannot be replaced by comparison counts.

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Basic search

Use zero-based node arrays. Find the first key≥k,or use key count if none.

```
search(x,k):
    i = first index with x.keys[i]>=k, or len(x.keys)
    if i<len(x.keys) and x.keys[i]==k: return (x,i)
    if x is leaf: return NOT_FOUND
    read page x.children[i]
    return search(x.children[i],k)
```

Without an equal key,the separators leave at most one child that can contain k. All others are excluded by global order. Remaining height decreases at each step,ending at a successful key or unsuccessful leaf. Duplicate checking before insertion adds at most one path and leaves the asymptotic I/O bound unchanged.

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Complete relocation when splitting a full child

Assume parent p is not full and y=p.children[i] has2t−1 keys. Save median=y.keys[t−1] before splitting.

```
z.keys = y.keys[t .. 2t-2]
y.keys = y.keys[0 .. t-2]
if y is internal:
    z.children = old y.children[t .. 2t-1]
    y.children = old y.children[0 .. t-1]
p.keys.insert(i,median)
p.children.insert(i+1,z)
write y,z,p
```

Each side has t−1 keys and,if internal,t children. The median moves out of the child into the parent as a real key; it is not duplicated. Original child ranges already lie on the correct sides of the median,so order and leaf depth remain. The nonfull parent can accept one key without exceeding2t−1.

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Split before descending to keep the current node nonfull

```
insert(k):
    if search(root,k) succeeds: return EXISTS
    if root is full:
        root = new empty internal node with old root as child0
        split(root,0)
    x=root
    while x is internal:
        i = child index selected by k
        if x.children[i] is full:
            split(x,i)
            if k>x.keys[i]: i++
        x=x.children[i]
    insert k into x.keys in sorted position
```

Invariant:x is nonfull. Splitting a full child before descent yields a chosen t−1-key child,so the next x is also nonfull. Leaf insertion adds at most one key without overflow. At most one split occurs per level. Only root splitting raises height,and it raises every leaf equally.

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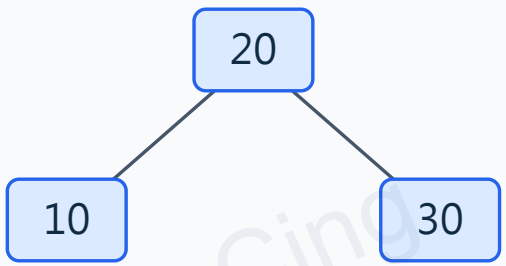
Split before descending to keep the current node nonfull (continued)

Split a full root, then insert25

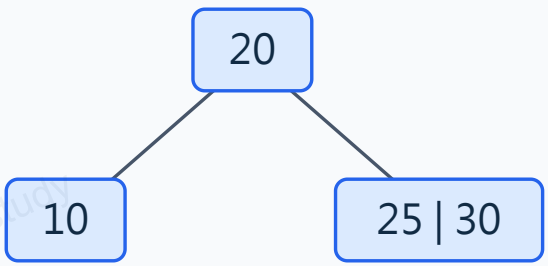
1 Before insertion, t=2



2 Split, before insert25



3 Insert25 on the right



Median20 moves up and leaves the old leaf. Since25>20, insert25 into the right leaf.

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Numerical root-split check

At t=2,insert25 into full leaf root[10,20,30].

Stage	Root	Children	Inorder
Before	[10,20,30],leaf	None	10,20,30
Split first	[20]	[10],[30]	10,20,30
Insert25	[20]	[10],[25,30]	10,20,25,30

20 appears only at root.25>20 chooses child1; that leaf grows from1 to2 keys,within maximum3. Both leaves have depth1.

A split writes old child,new child,and parent,a constant number of pages. State whether parent/old child are already cached. O(h) means linear in height,not an identical fixed read/write count for every insertion.

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Deleting a key

Use a stronger descent condition: every recursively entered nonroot starts with at least t keys,so losing one leaves at least t−1. At the end,a zero-key internal root may be replaced by its sole child.

When k is found:

1. In a leaf,erase it directly.
2. In an internal node,if its left child has at least t keys,replace k by that subtree's maximum predecessor and delete it there. Otherwise use the successor if the right child has at least t keys.
3. If both children have t−1 keys,merge left child,parent key k,and right child into2t−1 keys,then delete k in the merged node.

Erasing an internal key without adjusting children breaks key/child counts and separators. Predecessor/successor replacement also moves the associated record data.

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Repair the target child before descending

If x lacks k,search order selects child y. Descend directly if y has$\geq$t keys. Otherwise y has t−1:

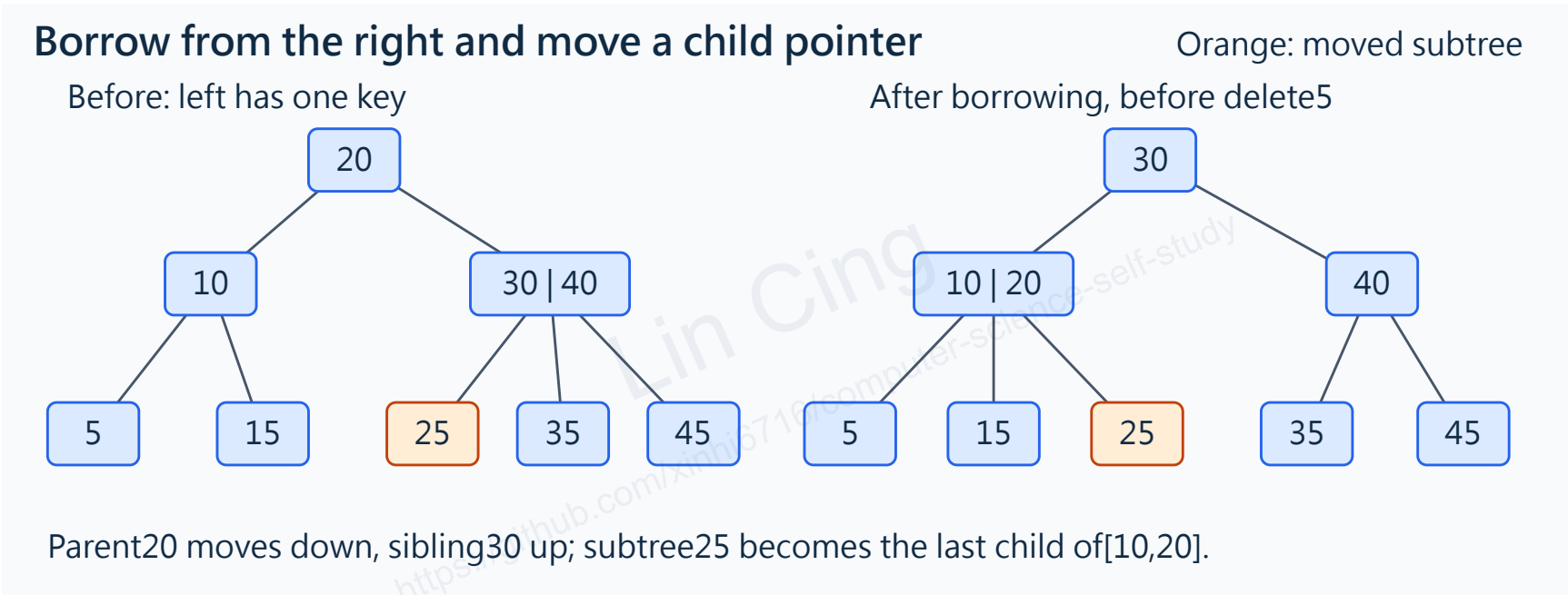
- With a left sibling holding$\geq$t,move the parent separator to y's front and the sibling's maximum into the parent. For internal nodes,move that sibling's rightmost child to y's first child position.
- Otherwise,with a rich right sibling,move the separator to y's end and that sibling's minimum into the parent. Move its leftmost child to y's last child position.
- Otherwise merge y with an adjacent sibling and their parent separator,yielding2t−1 keys. If merging left,reduce the target child index by1.

Borrowing preserves both children’ s lower bounds without changing parent key count. Merging removes one parent key and one child while preserving ranges and leaf depth. An absent k may still trigger legal restructuring before NOT_FOUND,but the key set must remain unchanged.

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Borrowing from the right also moves a child pointer

At t=2,deleting5 starts with a one-key left root child,too small for direct descent,and a two-key right child able to lend.



After borrowing,root30 has left child[10,20] with leaves[5],[15],[25],and right child[40] with[35],[45]. Subtree25 must move:left of new root30 and greater than20,it belongs as[10,20]'s last child.

Borrowing from the right also moves a child pointer (continued)

This is pre-descent repair;5 has not yet been removed. Next,[5] and adjacent[15] both have one key,so merge them with parent10 to[5,10,15],then delete5 to get[10,15]. Left internal node becomes[20],children[10,15],[25]. Root30 and right side remain. All leaves still have depth2.

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Internal replacement and root contraction

For root[20],left leaf[5,10],right leaf[30],delete20 via predecessor10. Result root[10],left[5],right[30],inorder5,10,30. Do not copy10 upward and forget to remove its old leaf occurrence.

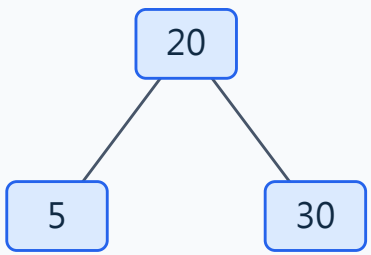
For root[20] with leaves[5],[30],both are minimum-size. Merge to[5,20,30],leaving the old root temporarily empty with one child. Delete20 in the merged leaf to obtain[5,30],then replace the empty internal root by that child.

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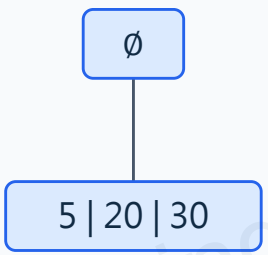
Internal replacement and root contraction (continued)

Merge, delete, and contract the root

1 Before delete20, t=2



2 Merge; old root empty



3 Delete20; contract root



Merge leaves and separator20; delete20 there; promote the sole child of the empty root.

Stable height1→0; only20 is removed from{5,20,30}.

Only an empty root may contract this way. Bypassing a general internal node would violate equal leaf depth. Deleting the last key retains a zero-key leaf root as the empty tree.

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Deletion pseudocode and induction

```
delete(x,k):
    i = first key >=k, or key count
    if key i equals k:
        if leaf: erase key i; return FOUND
        if child i has >=t keys: replace k by pred; delete(child i,pred)
        elif child i+1 has >=t keys: replace k by succ; delete(child i+1,succ)
        else: y=merge(child i,key i,child i+1); delete(y,k)
        return FOUND
    if leaf: return NOT_FOUND
    y=child i
    if y has t-1 keys: borrow from a rich neighbor, else merge as above
    return delete(the repaired target child,k)
after top call: if root has0keys and1child: root=its child
```

Induct on remaining height. The descent condition lets a leaf lose one key without falling below t−1. Internal replacement preserves order while removing exactly target k from the total set. Borrow/merge gives a legal smaller-height subproblem. A nonroot x starts with≥t keys,so losing one separator in a merge remains legal. Final root contraction handles the exception. This proves termination,key-set correctness,and structural invariants.

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Operation costs and implementation checks

Each descent handles constantly many adjacent nodes,with O(1) page reads/writes per level. Search,insertion,and deletion use O(h+1)=O(1+log_t(n+1)) page accesses. Within-node key/child movement costs O(t),giving CPU O(t(h+1)). Finding predecessor/successor separately before deletion adds only a constant number of paths in that subtree; these can also be combined into one descent.

Assume each node fits a page and a fixed root-cache model. Concurrency locks,transaction logs,compression,and fragmentation are additional implementation contracts,not consequences of abstract B-trees.

After each operation,check global key ranges,nonroot occupancy,child count=keys+1,equal leaf depth,no repeated child pointers,and inorder agreement with a reference set. Finite programs catch implementation errors; the induction proves arbitrary legal operations.

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Sources

Bibliography

Cormen, Leiserson, Rivest, Stein. *Introduction to Algorithms*, 3rd ed., MIT Press, 2009, Chapter18.

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References and attribution

Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein. Introduction to Algorithms, 3rd Edition.

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